

Shivanand Verma

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Professional Summary

Motivated CS undergraduate specializing in **AI/ML** with strong hands-on experience building **full-stack web applications**, **deep learning models**, and **AI-powered tools**. Built hybrid CNN achieving **93%+ F1-score** for medical image classification; developed AI code vulnerability scanner with **94% accuracy**. Solved **600+ problems** on LeetCode; secured **International Rank 1** in AIST Contest. Proficient in **Python, Java, JavaScript, TensorFlow, React**, and **SQL** with solid foundations in **DSA, DBMS**, and **System Design**.

Education

Pranveer Singh Institute of Technology — CGPA: 8.1/10

2022 – 2026

B.Tech in Computer Science & Engineering (AI/ML)

Technical Skills

Languages: Python, Java, C++, JavaScript (ES6+), SQL, PHP

AI/ML: TensorFlow, Keras, Scikit-learn, Deep Learning, Transfer Learning, Grad-CAM, Computer Vision

Web & Frameworks: React.js, FastAPI, HTML5, CSS3, RESTful APIs

Databases: MySQL, MongoDB, Database Design, Query Optimization

Tools & Deployment: Git, GitHub, VS Code, Vercel, Render, Railway, CI/CD

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Projects

◇ **Hybrid Skin Cancer Detection with Explainable AI** [\[GitHub\]](#)

Feb 2025 – Mar 2026

Python, TensorFlow, Keras, Scikit-learn, Grad-CAM

- Engineered hybrid ensemble CNN combining **5 architectures** (VGG, ResNet, DenseNet, Xception, MobileNet) for 7-class skin lesion classification on **10,000+** dermoscopic images, achieving **93%+ F1-score**.
- Implemented **Grad-CAM** heatmap visualizations to surface pixel-level model reasoning, increasing prediction interpretability by **70%** and enabling clinician-facing explainability.
- Applied **transfer learning** and **8 augmentation techniques**, expanding effective dataset size by **5×** and reducing model overfitting by **22%** compared to baseline single-architecture models.
- Optimized evaluation pipeline using **AUC, F1-score**, and threshold tuning across all 7 classes, achieving **91%+** precision with robust multi-class classification performance.

◇ **AIX-Scanner: AI Code Vulnerability Scanner** [\[GitHub\]](#)

Apr 2025 – May 2025

React.js, FastAPI, CodeBERT, MongoDB

- Built hybrid detection engine combining fine-tuned **CodeBERT** with rule-based static analysis, identifying **26+** vulnerabilities across **10 CWE categories** with **94%** accuracy.
- Developed auto-remediation system generating secure code replacements with one-click bulk fixing, reducing manual fix time by **85%**.
- Shipped full-stack app with **Monaco Editor**, glassmorphism UI, real-time risk scoring, and **PDF reports** — deployed via **Vercel + Railway** with CI/CD.
- Designed **MongoDB** schema for storing scan history, vulnerability logs, and user sessions, supporting efficient retrieval of **10,000+** scan records.

◇ **Lost and Found System** [\[GitHub\]](#)

Feb 2023 – Dec 2023

PHP, MySQL, JavaScript, HTML5, CSS3

- Developed a full-stack web application with **2 modules** (Admin and Public) to manage lost and found item reporting, tracking, and claim processing for **500+** users.
- Built responsive admin dashboard with **CRUD operations** using **PHP** and **MySQL**, managing **1,000+** item records with role-based access control.
- Implemented dynamic search and category-based filtering, reducing average item lookup time by **60%** for public users.
- Designed normalized **MySQL database schema** with **8+** relational tables for users, items, categories, and claim status tracking with optimized query performance.

Achievements & Certifications

- AIST Contest – International Rank 1 (2024 & 2025):** Secured 1st place consecutively among global participants; awarded **\$6,000** total prize money across 2 editions. [\[Link\]](#)
- Amazon ML Summer School 2025:** Selected among top applicants nationwide; trained in ML, NLP, Computer Vision, and RL by Amazon scientists.
- Competitive Programming:** Solved 600+ problems on [LeetCode](#); HackerRank: 5-Star Problem Solving, 5-Star C++.
- Coursera – Stanford University:** Completed Unsupervised Learning, Recommenders, and Reinforcement Learning. [\[Link\]](#)
- Salesforce Trailhead:** Adventurer & Agentblazer Champion Badge Holder – expertise in Salesforce AI Agent ecosystem and CRM automation. [\[Link\]](#)